

Tasks

Interests:

- **Diversity > Quantity**; We are looking for diversity in manipulation skills, tasks, and scenes.
- Bimanual coordinated tasks
- Longer horizon tasks (these can be broken down into sub-tasks)
- High precision / more contact-rich tasks

Tasks:

1. **Pick, Place & Reorient Objects**
 - a. **Pick & Place** grocery items from/to shelves, tables, and totes.
 - b. **Emptying/Filling a tote / box / bag** of sparse, clustered, and tessellated (similar or different) items.
 - c. **Packaging and Kitting** (see toy example, etc)
 - d. **Sorting** (trash / non-trash item)
2. **Assembly and Fastening** (require **precision**, **bimanual coordination**, and **compliance**)
 1. **Opening and closing** (e.g. a box, a case, a bag, door, closet, ...)
 2. **Insertion / Stacking** (e.g. stacking big legos, insert rings along a bar, ...)

Examples using kid toys:



For the Kitting toy, here is a possible task scenario:

- All items are outside the container on a table.
- The robot needs to **pick** and **place** the items in specific locations in the container.
 - For some items, the robot needs to carry out extra manipulation skills. E.g. for the bottle, it might need to **screw** the cap. For the towel, it might need to **fold** it.
- The robot needs to **open** a closet.
- The robot then **picks** the container and **places** it on a shelf.

There are many other task scenarios that can be carried out.

Task variation

- the target item pose
- the main gripper → don't only use your right hand.
- the starting configuration
- the execution order
- the motion trajectory → this doesn't mean unnatural motions.
- Min amount: 50 demos.
- Avoid diversifying the velocities.

Scene variation

- Background scene: lighting, workspace, environments, etc.
- Items (including distractors): the types, amounts, shapes, colors, opacity, weight, poses.